Incompressible Turbomachines

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Fanial Project

Investigation of the Effect of Blade Number on the Head–Flow Rate Characteristic Curve of a Centrifugal Turbopump Using CFD

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Abstract

This project investigates the effect of impeller blade number on the performance of a centrifugal turbopump using Computational Fluid Dynamics (CFD). Three impeller configurations with 3, 5, and 7 blades were modeled and simulated in ANSYS Fluent at a rotational speed of 1450 rpm, with water as the working fluid. The pressure distribution, velocity field, convergence behavior, and outlet flow rate were evaluated for each configuration. The numerical results were also compared with the theoretical head–flow rate characteristic curve, and turbomachinery similarity relations were used to estimate the performance of geometrically similar pumps with larger impeller diameters. The results show that changing the blade number affects the internal flow structure, pressure rise, and outlet flow rate, while the CFD predictions remain in satisfactory agreement with the theoretical analysis.

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Introduction

Centrifugal pumps are among the most widely used turbomachines for transferring energy to fluids in industrial, commercial, and engineering applications. The hydraulic performance of a centrifugal pump is strongly influenced by the geometry of its impeller, particularly the number of blades. Variations in blade number can affect the flow structure inside the impeller passages, pressure rise, flow rate, hydraulic losses, and ultimately the overall pump performance.

The head–flow rate characteristic curve is one of the main parameters used to evaluate the performance of a centrifugal pump. Although theoretical relations can provide an initial prediction of pump behavior, the internal flow field is inherently complex due to effects such as turbulence, blade loading, circulation, slip, flow separation, and blockage caused by the finite thickness of the blades. Computational Fluid Dynamics (CFD) provides an effective approach for investigating these phenomena and evaluating the influence of geometric parameters on turbopump performance.

In the present project, the effect of the number of impeller blades on the performance of a radial centrifugal turbopump is investigated numerically using ANSYS Fluent. Three impeller configurations containing 3, 5, and 7 blades are considered while the main geometric characteristics of the blades and the rotational speed are kept unchanged. The pump operates at a rotational speed of 1450 rpm, and water is used as the working fluid.

The three-dimensional geometries are developed in ANSYS SpaceClaim, followed by mesh generation in ANSYS Meshing. The turbulent flow through the turbopump is then simulated in ANSYS Fluent using the k–ε turbulence model. For each blade configuration, the pressure and velocity distributions, numerical convergence behavior, and outlet flow rate are evaluated and compared.

In addition to the CFD analysis, theoretical calculations are performed to obtain the theoretical head–flow rate characteristic curve. The numerical results are compared with the theoretical prediction to assess the consistency of the simulation. Furthermore, turbomachinery similarity relations based on the flow coefficient and head coefficient are employed to investigate the scaling of the obtained results to pumps with larger impeller diameters.

The main objective of this study is therefore to determine how changing the number of impeller blades influences the flow characteristics and performance of the centrifugal turbopump and to examine the agreement between CFD predictions, theoretical calculations, and turbomachinery similarity relations.

Simulation in ANSYS Fluent

In this project, turbopumps with **3, 5, and 7 blades** were simulated. Initially, the geometry was modeled in the **SpaceClaim** environment. The blade specifications were kept identical for all three configurations. The blade specifications are as follows:

$R_1 = 7.5\ mm$, $R_2 = 25\ mm$, $\beta_1 = 7.5\ mm$, $\beta_2 = 7.5\ mm$, $b_1 = 0.3\ mm$, $b_2 = 2\ mm$

The rotational speed was also set to **1450 rpm**.

2-1 Implemented Geometry

The overall implemented geometry is shown below. Cross-sectional views corresponding to the different numbers of blades are also presented in the following subsections.

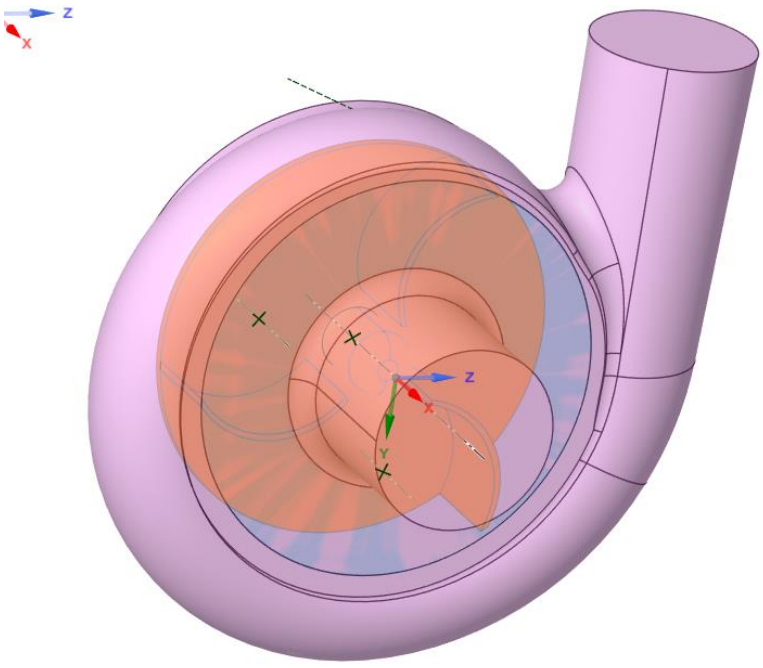


Figure 1: Overall view of the three-dimensional simulation of the radial turbopump in SpaceClaim

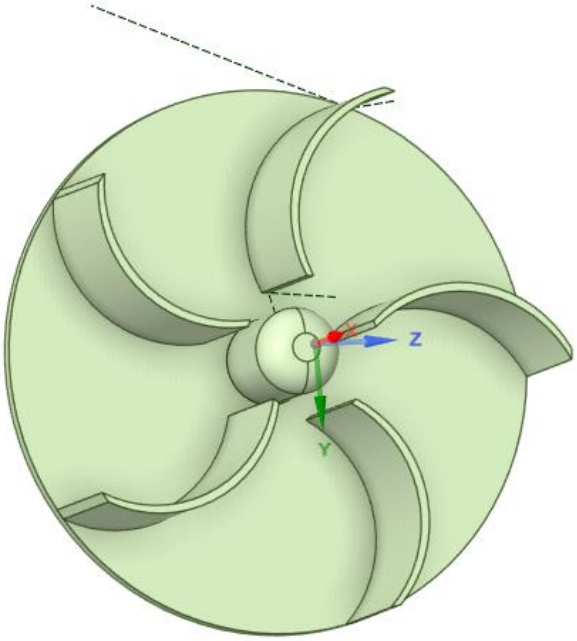


Figure 2: Example of the impeller modeled for the three-blade turbopump in SpaceClaim

2-1-1 Three-Blade Geometry

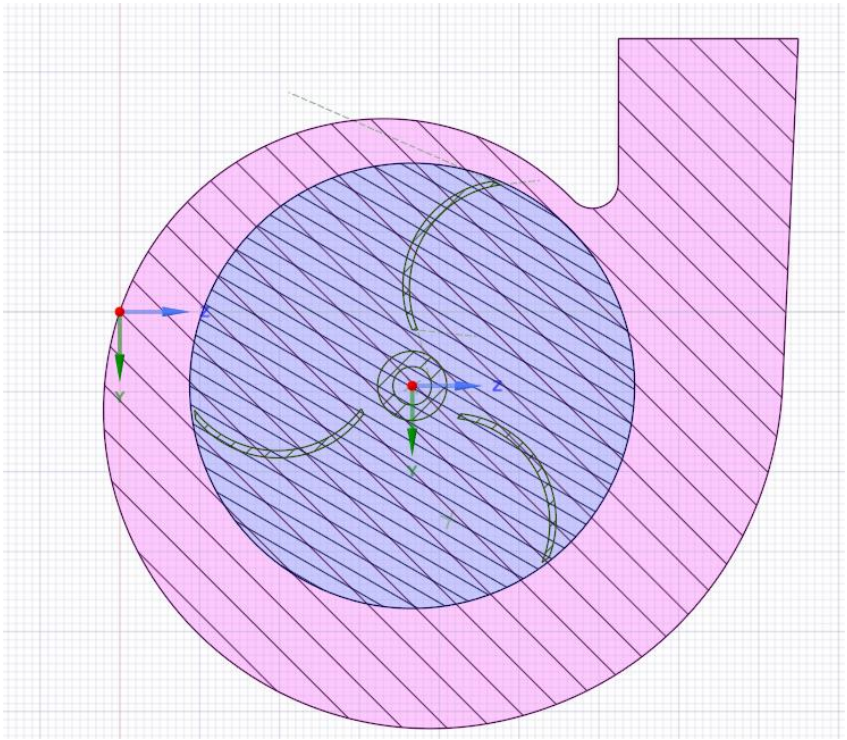


Figure 3: Cross-sectional view of the three-dimensional radial turbopump geometry for the three-blade configuration

2-1-2 Five-Blade Geometry

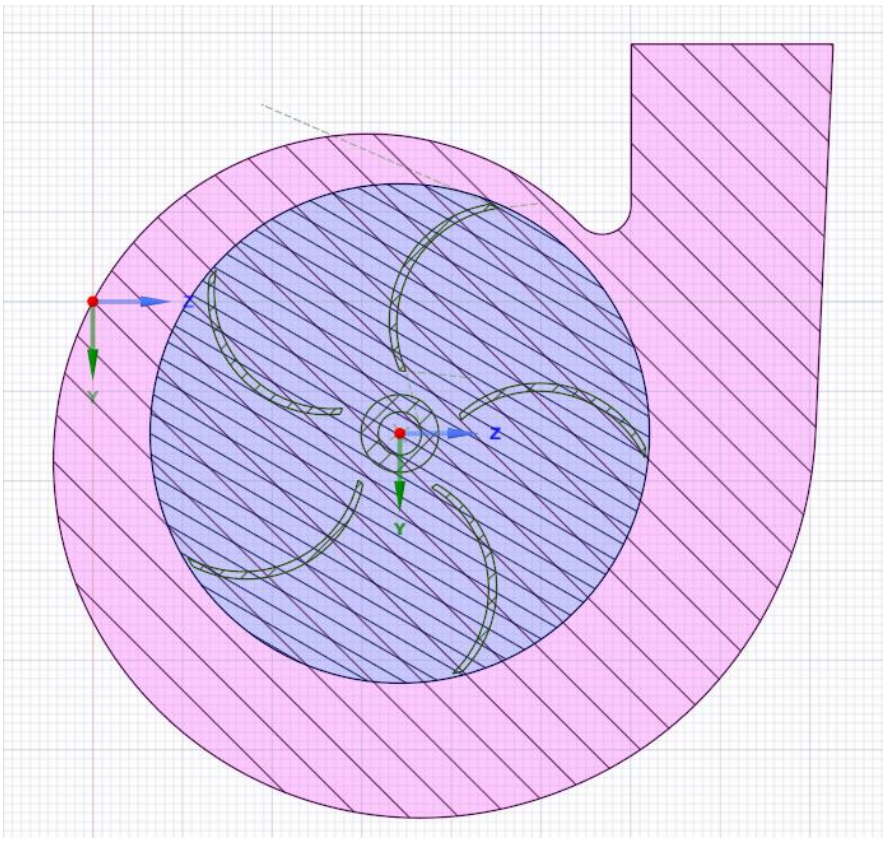


Figure 4: Cross-sectional view of the three-dimensional radial turbopump geometry for the three-blade configuration

2-1-3 Seven-Blade Geometry

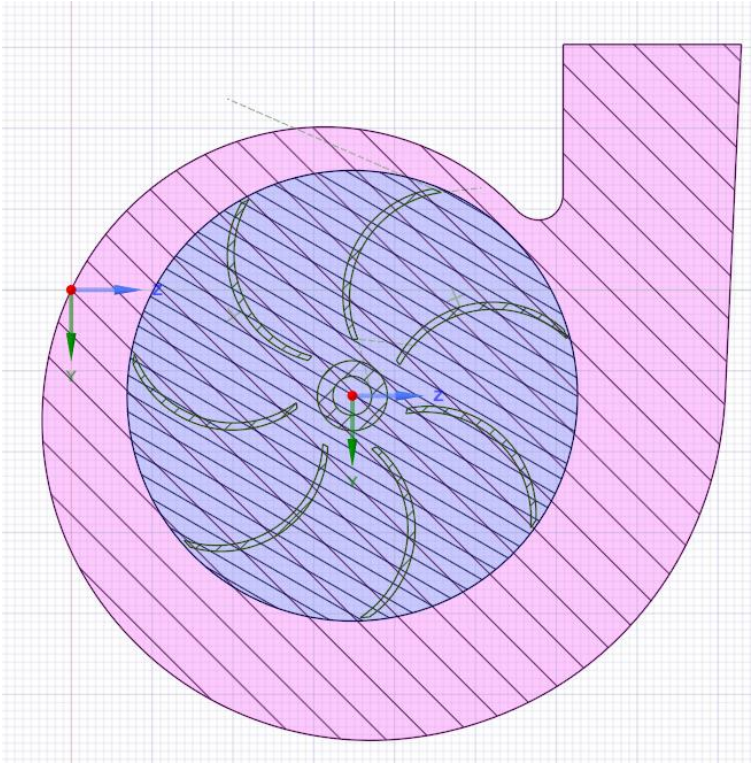


Figure 5: Cross-sectional view of the three-dimensional radial turbopump geometry for the seven-blade configuration

2-2 Mesh Generation

Next, a high-quality structured mesh containing approximately 640,000 elements was generated using ANSYS Meshing. The generated mesh for the three-blade turbopump is shown below.

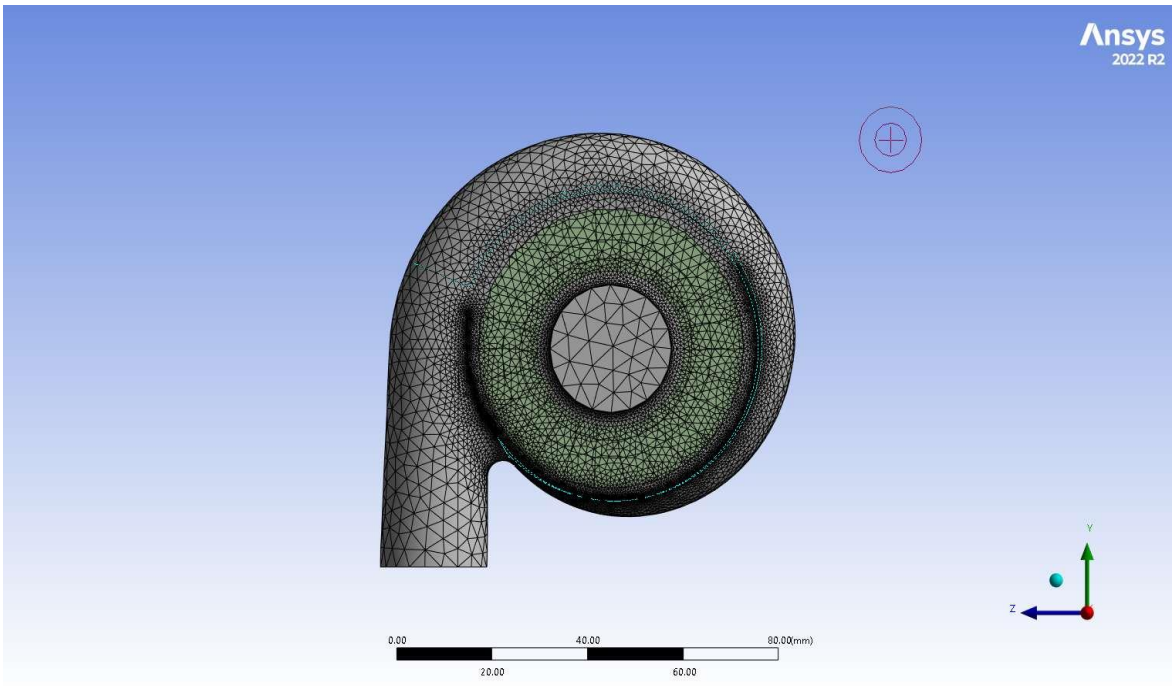


Figure 6: Generated mesh for the three-blade turbopump

2-3 Simulation Results

In the ANSYS Fluent environment, the numerical settings appropriate for the present application were applied. The Reynolds number of the turbopump indicates turbulent flow; therefore, the k–ε turbulence model, together with the standard wall function, was selected for predicting turbulent flow and near-wall regions.

To improve numerical accuracy, the Double Precision option was enabled so that the calculations would be performed with higher numerical precision. In addition, 7 CPU cores were used to reduce the computational time.

Water was selected as the working fluid, with its density and viscosity specified according to the values given in the original model.

The turbulence specification method was set to Intensity and Viscosity Ratio, with a Turbulent Intensity of 5% and a Turbulent Viscosity Ratio of 10.

The SIMPLEC pressure–velocity coupling algorithm was used, together with second-order discretization schemes.

In general, the selected convergence criterion is considered appropriate for turbulent flow simulations, and the same criterion was adopted in this project. At the end of the simulations, the

convergence requirements were satisfactorily achieved. The residual plots for each blade configuration are presented separately in the following sections.

2-3-1 Three-Blade Turbopump

A time step of **0.5 s** was used, and the total number of time steps was set to 100 to achieve satisfactory convergence.

The results showed that the velocity components satisfied the prescribed convergence criterion. However, as expected, the continuity, k, and ε residuals did not fully reach the specified criterion, although they approached it sufficiently closely. Therefore, residual limits of the order specified in the original simulation were considered for k, ε, and continuity.

The residual convergence history of the governing flow equations is shown below.

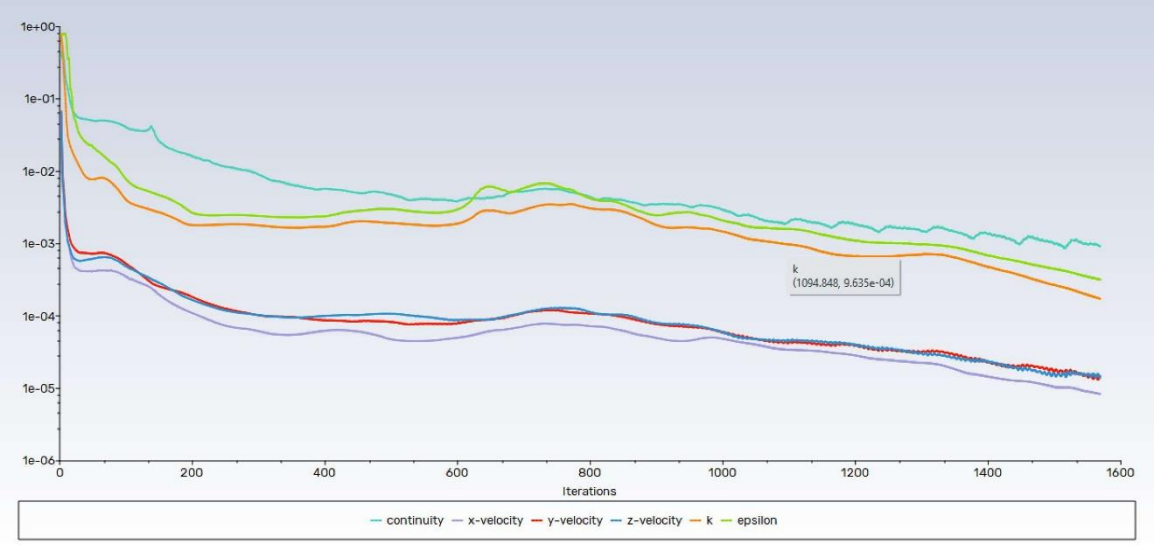


Figure 7: Residual convergence plot for the three-blade turbopump

The total pressure contour was subsequently obtained on the meridional plane of the pump. As expected, the pressure is lower near the blade inlet. After the fluid enters the pump, part of its kinetic energy is converted into pressure as it passes through the diverging flow passage, thereby contributing to pressure recovery.

The flow pressure at the outlet is higher than that at the inlet, which is consistent with the expected operating behavior of a pump.

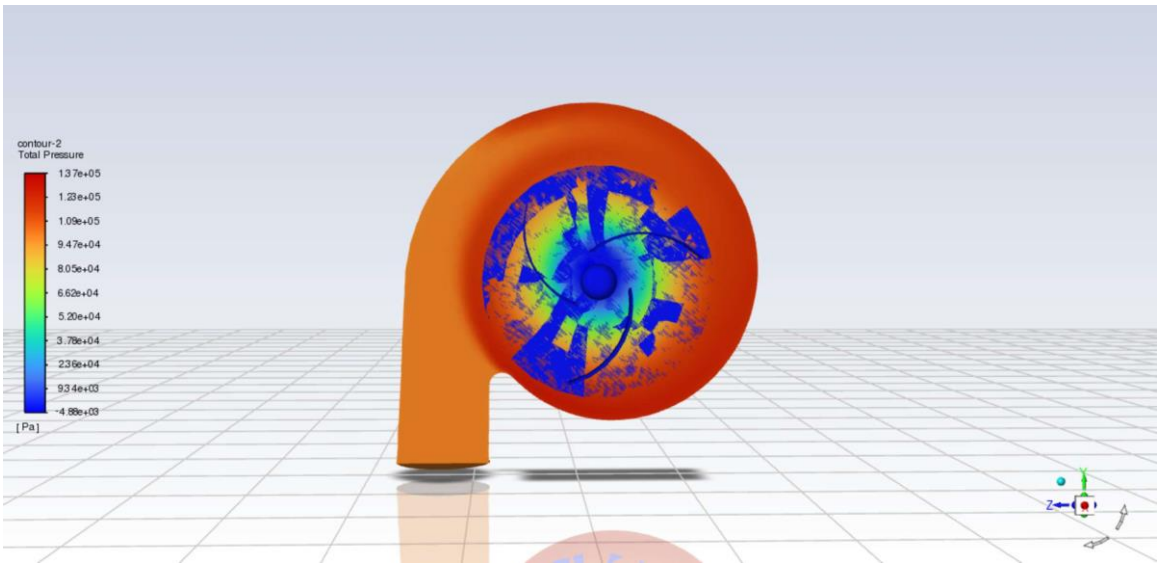


Figure 8: Total pressure contour on the meridional plane of the three-blade turbopump

For further investigation, the velocity contour on the same plane was also obtained.

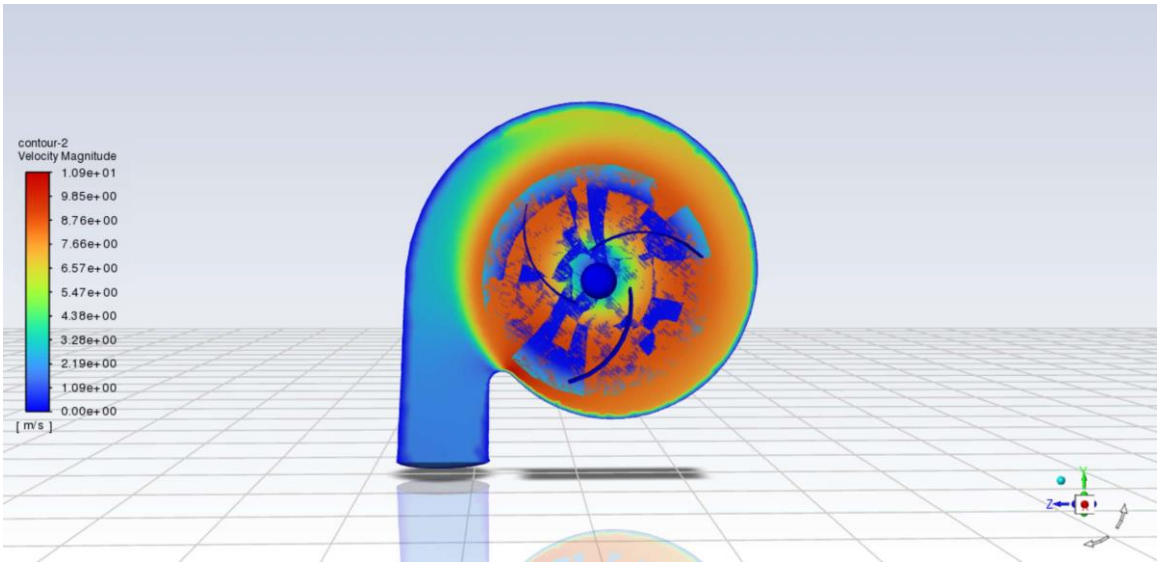


Figure 9: Velocity magnitude contour on the meridional plane of the three-blade turbopump

As expected, the velocity reaches its maximum value near the pump inlet. After the flow enters the passage between the blades, part of the fluid kinetic energy is converted into pressure. Consequently, the velocity decreases along the diverging outlet passage compared with its inlet value.

Finally, the outlet flow rate as a function of flow-solution time was obtained as follows.

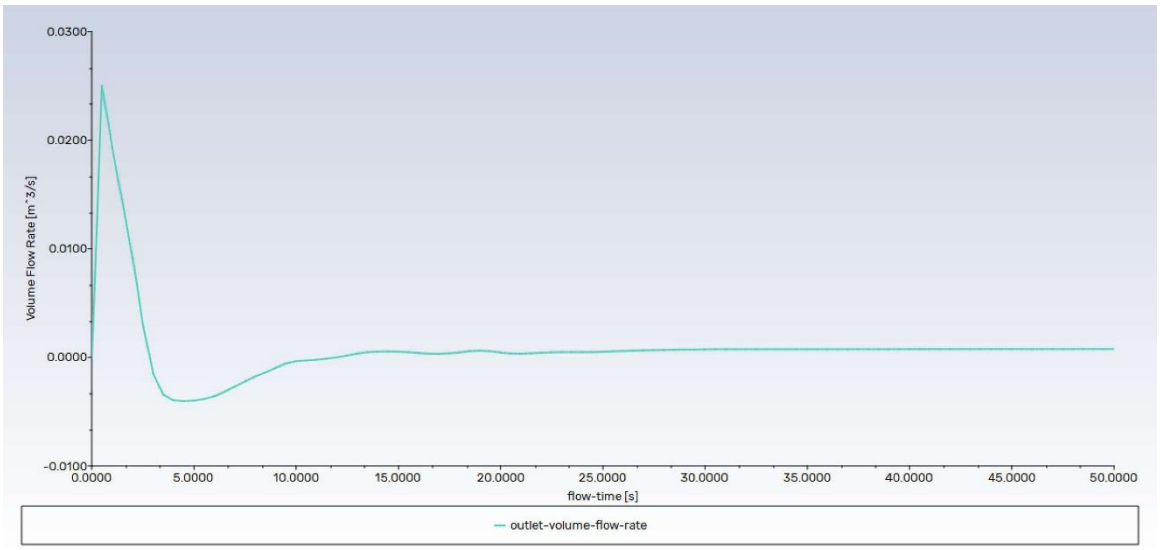


Figure 10: Outlet flow rate versus flow-solution time for the three-blade turbopump

At the beginning of the simulation, because the water enters in the form of a jet, reverse flow occurs and the calculated flow rate becomes negative. As the flow develops and approaches a stable condition, the water flow rate gradually converges to a specific value. In other words, the flow pattern inside the turbopump becomes established.

The final flow rate obtained for the three-blade turbopump was:

$$Q = 0.00074106694\ {}^{m^3}/_{s} = 0.741\ {}^{lit}/_{s}$$

2-3-2 Five-Blade Turbopump

The same numerical settings and convergence criteria used in the previous case were also applied to the five-blade configuration.

The residual convergence history of the governing flow equations is presented below.

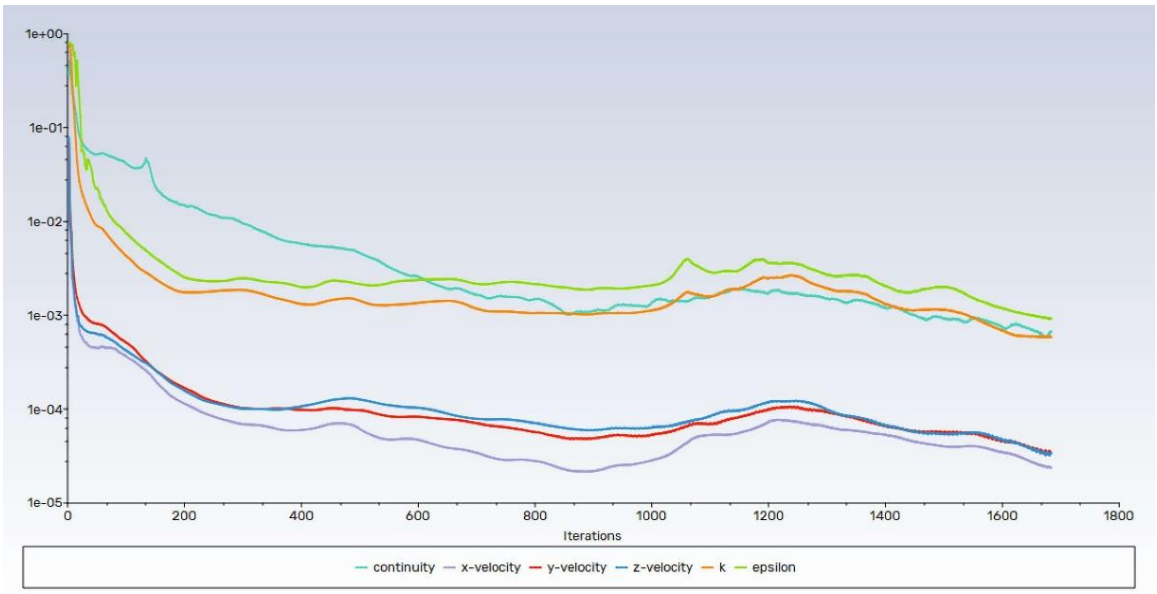


Figure 11: Residual convergence plot for the five-blade turbopump

The total pressure contour was then obtained on the meridional plane of the pump. Similar to the previous case, the pressure is lower at the blade inlet, and after the fluid enters the pump, part of its kinetic energy is converted into pressure along the diverging flow passage.

Compared with the three-blade turbopump, a greater pressure rise is observed in this configuration. The maximum pressure increases from the value obtained for the three-blade case to the corresponding value obtained for the five-blade case.

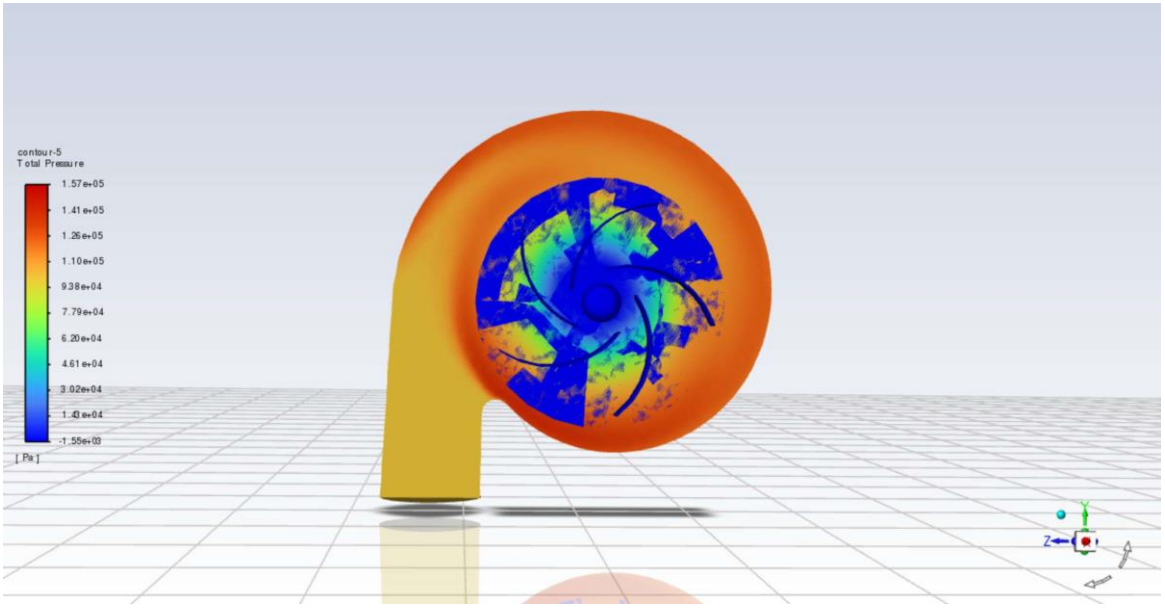


Figure 12: Total pressure contour on the meridional plane of the five-blade turbopump

The velocity contour for the five-blade turbopump was also obtained on the same plane.

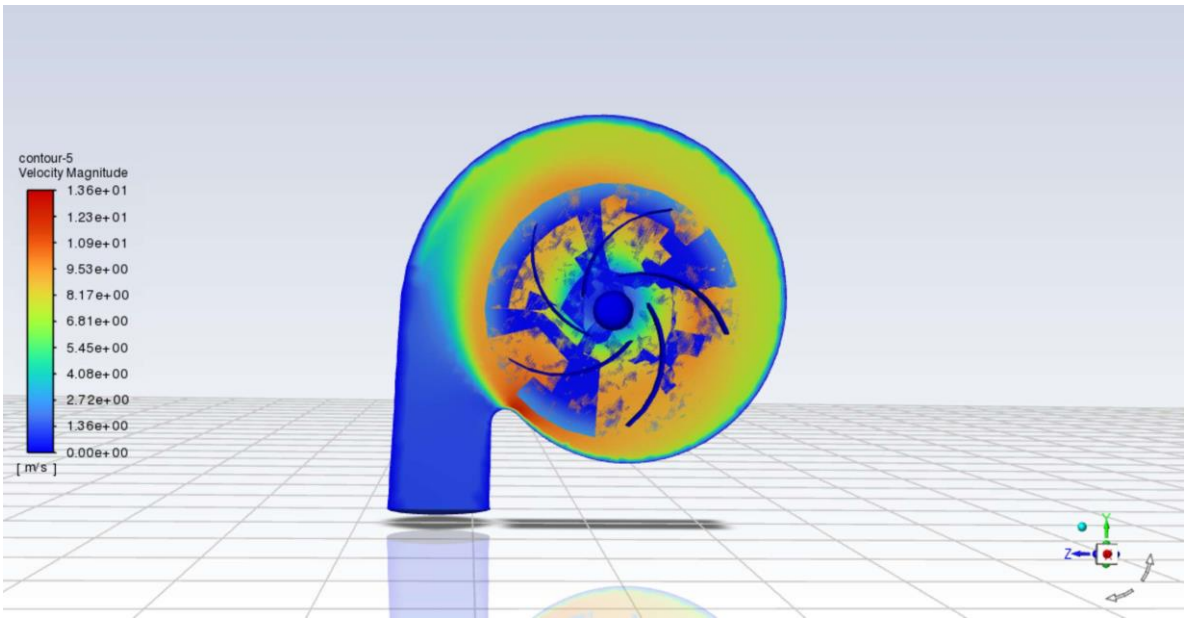


Figure 13: Velocity magnitude contour on the meridional plane of the five-blade turbopump

As expected, the velocity is highest near the pump inlet. After the fluid enters the passage between the blades, part of its kinetic energy is converted into pressure, and the velocity decreases along the diverging outlet passage relative to its inlet value.

The reduction in velocity is considerably greater than that observed for the three-blade turbopump. According to the interpretation adopted in this project, this indicates that the five-blade configuration produces lower losses than the three-blade configuration.

Finally, the outlet flow rate as a function of flow-solution time was obtained as follows.

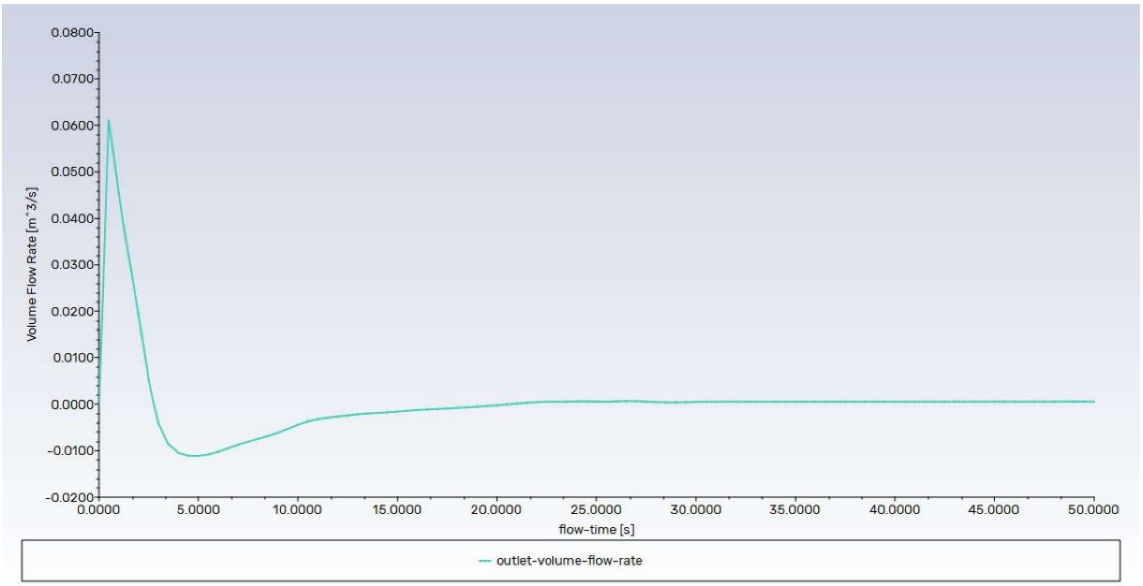


Figure 14: Outlet flow rate versus flow-solution time for the five-blade turbopump

Similar to the previous case, reverse flow initially occurs because the water enters as a jet, resulting in a negative flow rate. As the flow gradually stabilizes, the water flow rate converges to a specific value.

The final flow rate obtained for the five-blade turbopump was:

$$Q = 0.00056874975\ {}^{m^3}/_{s} = 0.569\ {}^{lit}/_{s}$$

2-3-3 Seven-Blade Turbopump

The same numerical settings and convergence criteria used in the previous cases were also applied to the seven-blade configuration.

The residual convergence history of the governing flow equations is presented below.

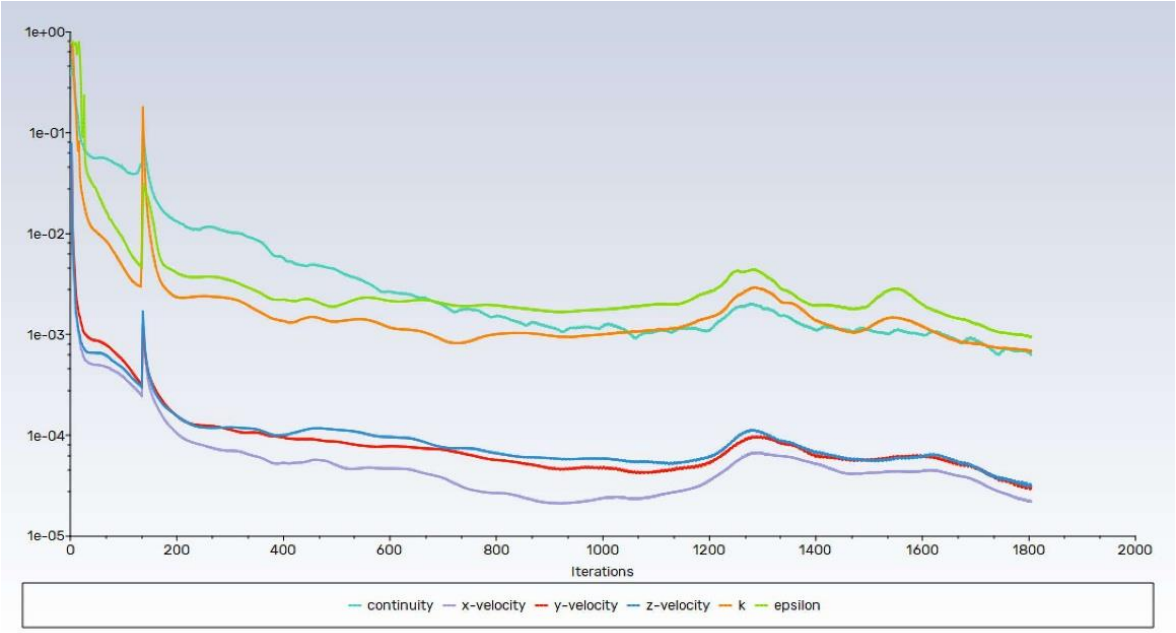


Figure 15: Residual convergence plot for the seven-blade turbopump

The total pressure contour was subsequently obtained on the meridional plane of the pump. Similar to the previous cases, the pressure is lower at the blade inlet, while part of the fluid kinetic energy is converted into pressure after entering the pump and passing through the diverging flow passage.

Compared with the five-blade turbopump, a greater pressure rise is observed. The maximum pressure increases from the value obtained for the five-blade configuration to that obtained for the seven-blade configuration.

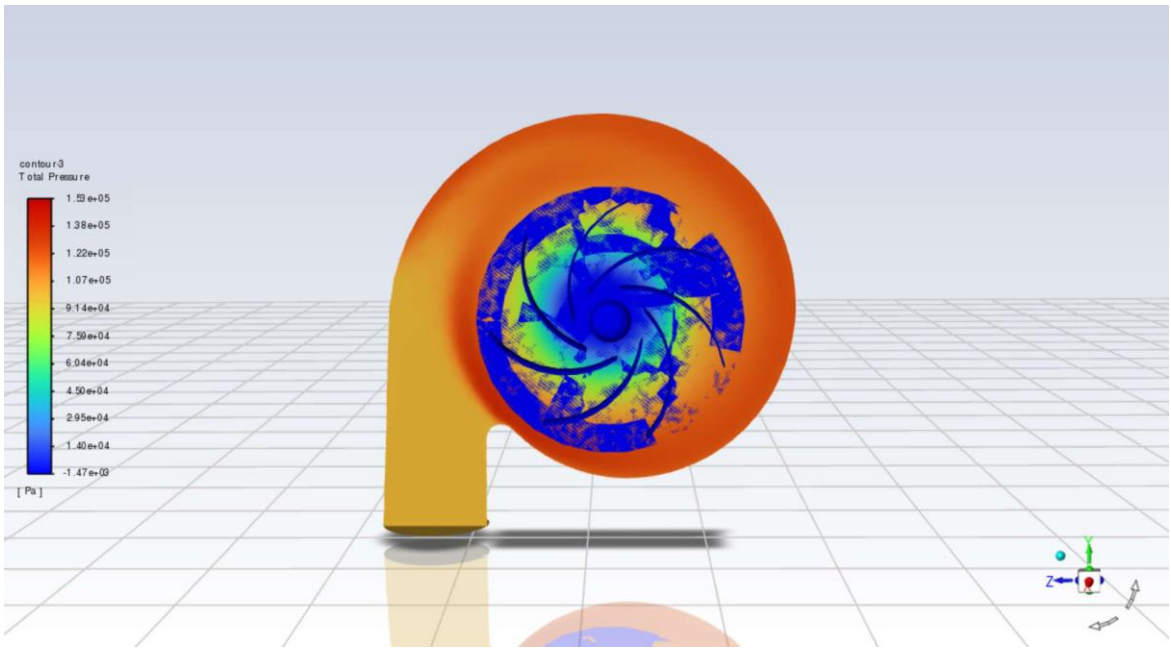


Figure 16: Total pressure contour on the meridional plane of the seven-blade turbopump

The velocity contour for the seven-blade turbopump was also obtained on the same plane.

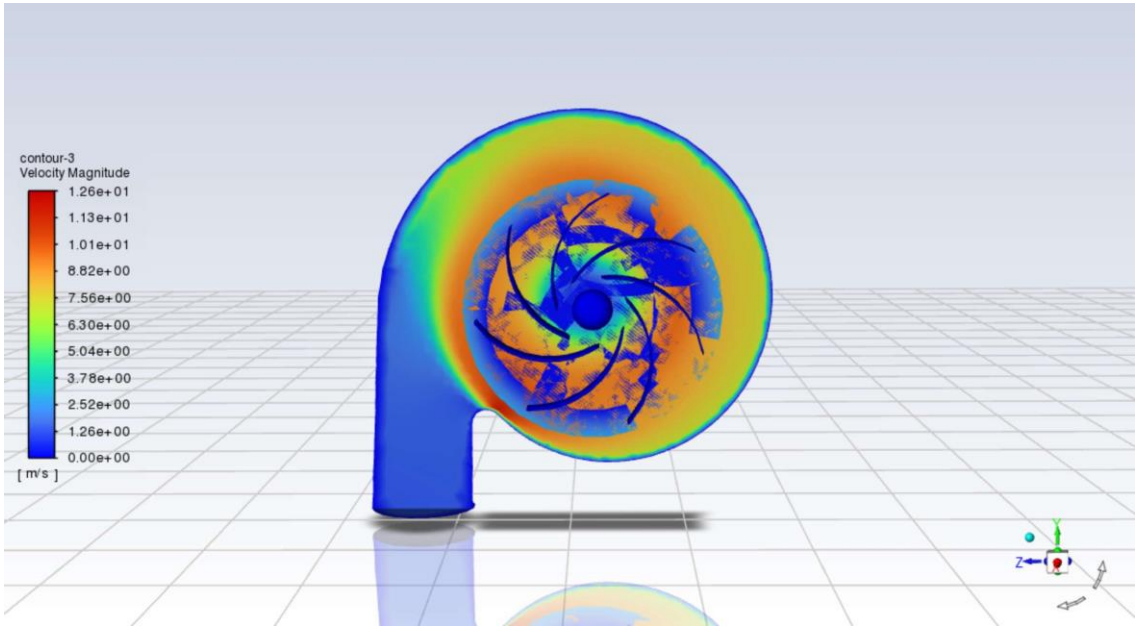


Figure 17: Velocity magnitude contour on the meridional plane of the seven-blade turbopump

As expected, the velocity is highest near the pump inlet and decreases along the diverging outlet passage.

The reduction in velocity is considerably greater than that observed in the five-blade turbopump. According to the interpretation adopted in this project, this indicates lower losses in the seven-blade configuration compared with the five-blade configuration.

However, within the blade-to-blade passages, the flow velocity is higher than in the five-blade configuration because increasing the number of blades reduces the available flow cross-sectional area.

Finally, the outlet flow rate as a function of flow-solution time was obtained as follows.

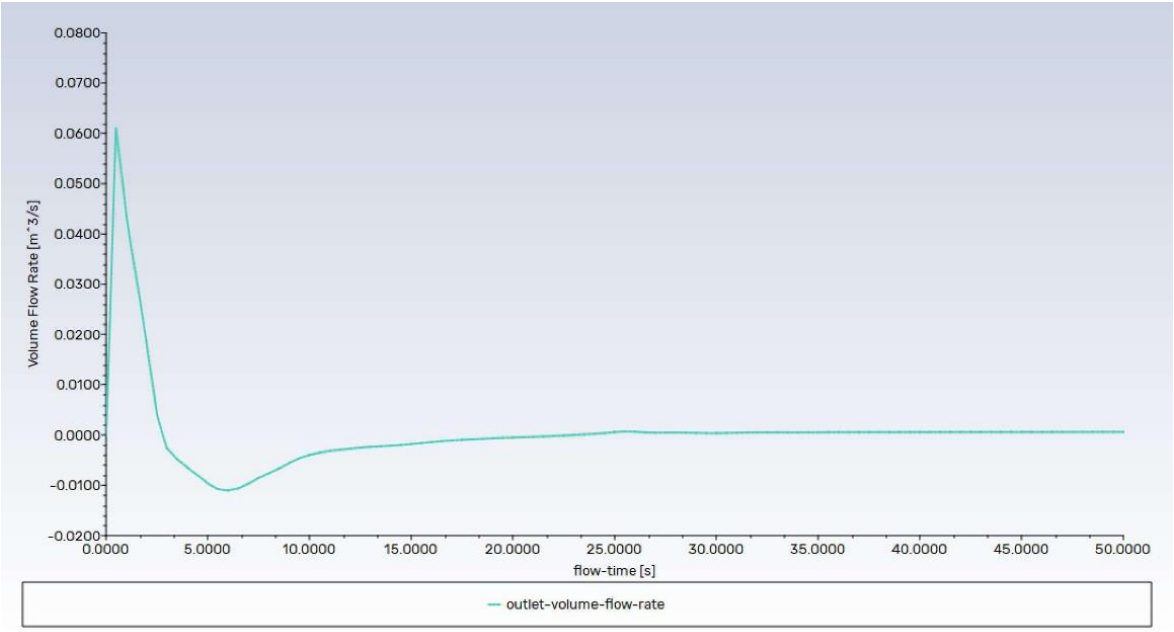


Figure 18: Outlet flow rate versus flow-solution time for the seven-blade turbopump

Similar to the previous cases, reverse flow initially occurs due to the jet-like entry of water, resulting in a negative flow rate. As the flow develops and reaches a stable condition, the water flow rate gradually converges to a specific value.

The final flow rate obtained at the end of the simulation was:

$$Q = 0.00050733971\ {}^{m^3}/_{s} = 0.507\ {}^{lit}/_{s}$$

Results Obtained from Theoretical Calculations

Considering the general relationship between pump head and flow rate, the theoretical head–flow rate characteristic curve can be obtained from the corresponding equation:

$$H = \frac{U_2 C_{u2} - U_1 C_{u1}}{g} \quad , \quad C_u = U - \frac{C_m}{\tan(\beta)} \quad , \quad C_m = \frac{Q}{A} = \frac{Q}{\pi D b}$$

$$\rightarrow H = \frac{U_2^2 - U_1^2}{g} + \left(\frac{U_1 \times \cot(\beta_1)}{\pi D_1 b_1 g} - \frac{U_2 \times \cot(\beta_2)}{\pi D_2 b_2 g} \right) Q$$

Therefore, using the blade geometric specifications presented in the previous section, the theoretical head–flow rate characteristic curve was obtained as follows.

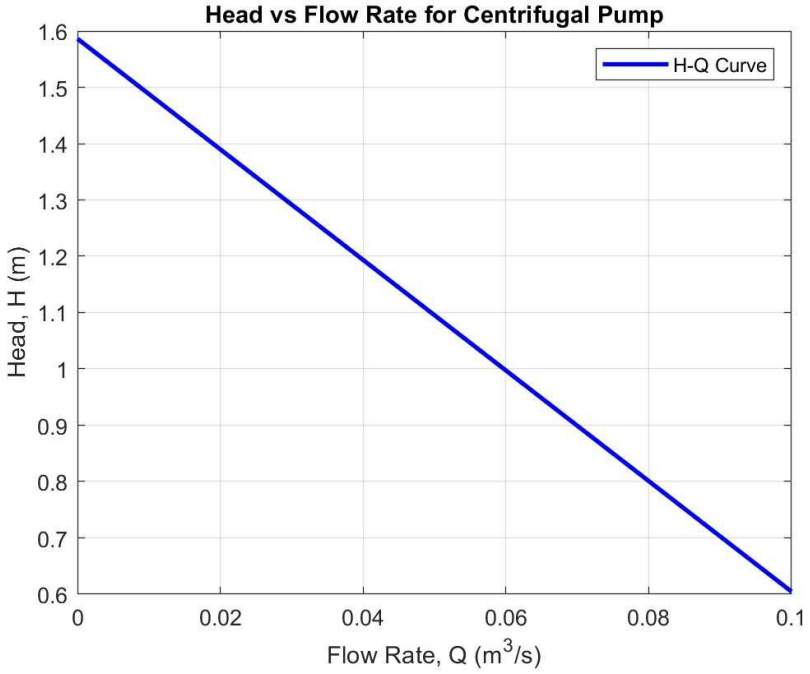


Figure 19: Theoretical head–flow rate characteristic curve of the turbopump

It can be observed that, for the generated head, the flow rate obtained from the turbopump simulation agrees closely with the theoretical head–flow rate characteristic curve. Therefore, the numerical results exhibit satisfactory agreement with the theoretical calculations.

Discussion and Analysis of Results

Increasing the number of blades reduces circulation and consequently decreases the associated head losses. Based on the outlet flow rates obtained from the simulations and their reduction with increasing blade number, this behavior was investigated.

On the other hand, reducing the number of blades increases the loading on each individual blade and consequently increases the pressure difference between the two sides of the blade. This produces a circumferential pressure gradient from the **pressure side** toward the **suction side**, which can contribute to the formation of the **jet–wake phenomenon**.

However, the simulation results indicated that the increase in blade loading caused by reducing the number of blades did not significantly affect the **slip**, and the resulting variations in flow rate and head remained relatively limited.

Furthermore, when complete similarity conditions are satisfied, geometrically similar turbopumps may be regarded as members of the same turbomachinery family. Therefore, the results obtained for the present geometry can be extended to larger turbopumps using the corresponding **flow coefficient** and **head coefficient**.

As indicated by the turbomachinery similarity relations, the pump head is proportional to the square of the impeller diameter, whereas the flow rate is proportional to the cube of the impeller diameter when the rotational speed and similarity conditions are appropriately considered.

4-1 Pump Similarity Analysis

The pump simulated in this project has an impeller diameter of **50 mm**. A relatively small impeller was selected in order to reduce the required mesh size and accelerate numerical convergence.

Using turbomachinery similarity relations, the obtained results can subsequently be scaled to estimate the outlet flow rates of larger pumps, including pumps with dimensions closer to conventional domestic applications.

4-1-1 Flow Coefficient

$$\delta = \frac{Q}{ND^3}$$

By keeping the rotational speed constant, the relationship between the flow rates corresponding to the original and modified impeller diameters can be determined.

$$\frac{Q_1}{D_1^3} = \frac{Q_1}{D_1^3}$$

$$Q_2 = Q_1 \times \left(\frac{D_2}{D_1}\right)^3$$

Accordingly, to calculate the flow rate for the new configuration, the original flow rate must be multiplied by the cube of the impeller-diameter ratio.

For this purpose, the new impeller diameter was assumed to be 100 mm. Therefore:

$$Q_2 = Q_1 \times \left(\frac{100}{50}\right)^3$$
$$=> Q_2 = Q_1 \times 8$$

Since the impeller diameter increases from 50 mm to 100 mm, the initial flow rate must be multiplied by 8 to obtain the corresponding flow rate for the larger pump.

Table 1: Effect of Impeller Blade Number and Diameter on Pump Outlet Flow Rate

Number of Impeller Blades	Impeller Diameter (mm)	Pump Outlet Flow Rate
3 blades	50	0.741
3 blades	100	5.928
5 blades	50	0.569
5 blades	100	4.552
7 blades	50	0.507
7 blades	100	4.056

As can be observed, increasing the number of pump blades reduces the outlet flow rate. This occurs because of the finite blade thickness at the impeller inlet.

Although increasing the number of blades produces a more streamlined flow pattern, and one might therefore expect the flow rate to increase, the simulations show a reduction in outlet flow rate. Thus, in the present geometry, increasing the number of blades has a negative effect on the delivered flow rate.

In theoretical analyses, blade thickness is often assumed to be negligible. Under such an assumption, increasing the number of blades reduces the losses associated with the pressure difference between the suction side and the pressure side, reduces circulation-related effects, and causes the outlet flow rate to approach the theoretical prediction.

In the present simulations, however, increasing the number of blades also increases the blockage at the impeller inlet. The reduced effective inlet flow area restricts the entry of fluid into the blade passages and consequently decreases the delivered flow rate.

4-1-2 Head Coefficient

$$\psi = \frac{gh}{ND^2}$$

Similarly, by keeping the rotational speed constant, the corresponding head similarity relationship can be written as:

$$\psi_1 = \psi_2$$
$$\frac{H_1}{D_1^2} = \frac{H_2}{D_2^2}$$
$$=> H_2 = H_1 \times \left(\frac{D_2}{D_1}\right)^2$$

As in the flow-rate analysis, the original impeller diameter is 50 mm, while the new impeller diameter is assumed to be 100 mm.

$$=> H_2 = H_1 \times \left(\frac{100}{50}\right)^3 = H_1 \times 4$$

Therefore, the original pump head must be multiplied by 4.

Since the simulated pump head was 10 m, the corresponding head for a geometrically similar pump with an impeller diameter of 100 mm is calculated as:

$$H = 40 \text{ m}$$

Thus, according to the turbomachinery similarity relations, doubling the impeller diameter while maintaining the stated similarity assumptions results in a fourfold increase in pump head.

Appendix A: MATLAB Code for Plotting the Theoretical Head–Flow Rate Curve

```
clc,clear;


Beta1 = 20.8;
Beta2 = 38;
D1 = 0.015;
D2 = 0.050;
b1 = 0.0003;
b2 = 0.002;
Omega = 1450 * 2 * pi / 60;
U1 = Omega * D1 / 2;
U2 = Omega * D2 / 2;
g = 9.81;
a = ((U1*cotd(Beta1))./(pi*D1*b1*g) - (U2*cotd(Beta2))./(pi*D2*b2*g));
b = (U2 ^ 2 - U1 ^ 2)/g;
Q = linspace(0, 0.1, 100); % Flow rate range (m^3/s)
H = b + a .* Q;


figure;
plot(Q, H, 'b-', 'LineWidth', 2);
grid on;
xlabel('Flow Rate, Q (m^3/s)');
ylabel('Head, H (m)');
title('Head vs Flow Rate for Centrifugal Pump');
   legend('H-Q Curve')
```